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Intrusion Detection and Prevention in MQTT-Based IoT Networks: A Comprehensive Review

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***Abstract*--The Internet of Things (IoT) has transformed sectors such as healthcare, transportation, and smart cities by enabling large-scale device connectivity. Among existing IoT communication protocols, Message Queuing Telemetry Transport (MQTT) is widely deployed due to its lightweight architecture and low overhead. However, its limited authentication and encryption features expose MQTT-enabled systems to significant security threats. This paper presents a comprehensive review of recent Intrusion Detection and Prevention Systems (IDS/IPS) developed for MQTT-based IoT networks, with a focus on machine learning, deep learning, hybrid, and federated learning approaches. MQTT-specific datasets, including MQTTset, DoS/DDoS-MQTT-IoT, and MQTT-IoT-IDS2020, are analyzed and compared with general IoT security datasets such as UNSW-NB15 and CICIDS2017. The review identifies key research challenges, including dataset quality constraints, scalability limitations, and insufficient model interpretability. The findings highlight the necessity for lightweight, adaptive, and explainable IDS/IPS frameworks capable of providing real-time protection in resource-constrained IoT environments. These observations motivate the development of the hybrid and interpretable security model proposed in this work.**

***Keywords- MQTT, IoT, IDS, IPS, Machine Learning, Deep Learning*.**

In the modern digital era, automation is rapidly advancing across sectors such as healthcare, transportation, manufacturing, and smart cities through interconnected devices that enable efficient data exchange and intelligent communication. These devices from household appliances to industrial sensors that are redefining system operations by supporting real-time monitoring, automated control, and data-driven decision-making. However, this large-scale expansion of connected technologies also brings significant challenges in security, scalability, and communication efficiency, highlighting the increasing complexity and vulnerability of the IoT ecosystem.

The IoT interconnects billions of devices that autonomously collect, exchange, and process data, enabling automation, remote monitoring, and intelligent decision-making across multiple domains. However, this rapid expansion has significantly widened the attack surface, exposing networks to risks such as unauthorized access, data manipulation, and distributed denial-of-service (DDoS) attacks. These challenges are further intensified by the limited processing power, memory, and energy available on many IoT devices, which restrict the deployment of traditional security solutions and underscore the need for lightweight yet effective protection mechanisms.

I. INTRODUCTION

The Internet of Things (IoT) has reshaped modern digital ecosystems by enabling seamless communication among interconnected devices in sectors such as healthcare, transportation, manufacturing, and smart cities [13]. Among the various communication protocols supporting IoT connectivity, the Message Queuing Telemetry Transport (MQTT) protocol is widely adopted for its lightweight design, low bandwidth usage, and ease of deployment [1]. MQTT operates on a publish–subscribe model that decouples message producers from consumers through a central broker. As shown in Fig. 1, devices publish messages to designated topics managed by the broker, which then forwards them to subscribed clients, enabling asynchronous and efficient data exchange suitable for low-power and bandwidth-constrained environments.[3].

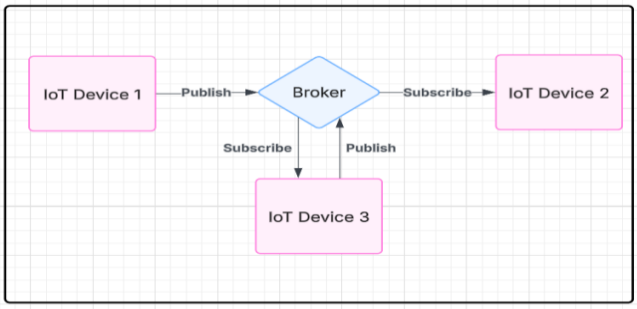


Fig 1. Basic MQTT Publish–Subscribe Communication Model.

Although MQTT's lightweight design supports widespread adoption, its default specification does not enforce encryption, authentication, or access-control measures, leaving deployments exposed to threats such as spoofing, replay attacks, man-in-the-middle (MitM) attacks, and denial-of-service (DoS) attempts [7]. The broker's central role further creates a single point of failure, increasing the impact of successful intrusions [11]. These weaknesses highlight the need for dedicated security mechanisms particularly Intrusion Detection Systems (IDS) and Intrusion Prevention Systems (IPS) to detect and mitigate malicious activities in MQTT-based IoT environments. [13].

Traditional security approaches, such as firewalls and antivirus software, are insufficient for distributed IoT environments composed of heterogeneous and resource-constrained devices [18]. IDS solutions monitor network traffic to identify anomalies or suspicious behaviors, while IPS mechanisms extend this capability by actively blocking or mitigating detected threats [9]. In MQTT-based IoT systems, a single compromised device or broker can rapidly spread attacks across the network, underscoring the need for intelligent and adaptive IDS/IPS frameworks. [5].

Recent studies have explored a range of IDS/IPS strategies, including signature-based, anomaly-based, and hybrid detection methods [15]. Signature-based techniques match traffic against known attack patterns and are effective for recognized threats but ineffective against zero-day attacks [16]. Anomaly-based approaches model normal network behavior to identify deviations, enabling detection of

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previously unseen attacks while often suffering from higher false-positive rates [17]. Hybrid methods integrate both techniques to leverage their respective strengths and improve overall detection accuracy and reliability [6].

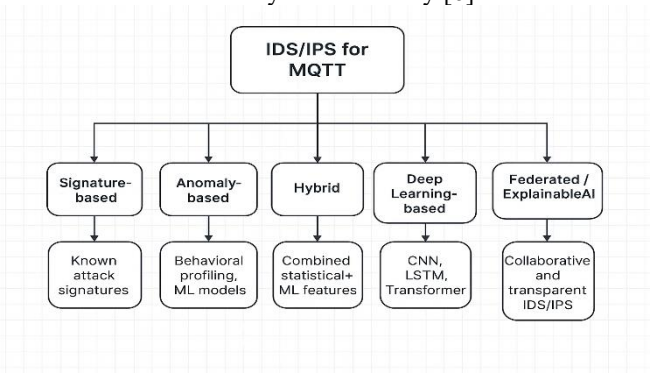


Fig 2. IDS/IPS Taxonomy for MQTT-Based IoT.

Advances in machine learning (ML) and deep learning (DL) have significantly enhanced intrusion detection by enabling automated feature extraction, pattern recognition, and adaptive learning from network traffic [3]. Recent IDS/IPS frameworks also incorporate emerging techniques such as federated learning, which supports collaborative model training without direct data sharing [10], and explainable AI, which improves transparency and operator trust [8]. The effectiveness of these systems is commonly assessed through metrics such as accuracy, precision, recall, F1-score, detection rate, false positive rate (FPR), and response time i.e, key indicators for real-time IoT security applications. [2].

The primary objective of this review is to provide a comprehensive analysis of existing intrusion detection and prevention mechanisms designed for MQTT-based IoT networks [4]. The study classifies detection approaches including ML, DL, statistical, signature-based, and hybrid methods that examines relevant datasets and evaluation metrics, and identifies key research challenges such as the lack of standardized datasets, limitations in real-time detection, scalability constraints, and the need for improved model explainability. [27].

The remainder of this paper is structured as follows. Section II presents a detailed literature review of IDS/IPS techniques applied to MQTT-based IoT environments, outlining their methodologies, strengths, and limitations. Section III discusses key challenges, open issues, and future research directions for securing MQTT-based IoT networks. Section IV concludes the paper by summarizing the main insights and highlighting potential pathways for advancing secure MQTT communication.

II. RELATED WORK

Research on IDS/IPS solutions tailored for MQTT has evolved along several complementary directions, including dataset development, protocol-aware preprocessing, classical machine learning, deep learning, hybrid models, federated or distributed architectures, and explainable system designs. The following sections synthesize representative contributions and key findings from these efforts.

Vaccari et al. [1] introduced MQTTset, one of the first labeled MQTT datasets, validating ML algorithms on protocol-aware features and providing a protocol-specific benchmark. Hindy et al. [2] released MQTT-IoT-IDS2020 and highlighted the value of flow-based feature representations in improving detection accuracy versus packet-only features. Khan et al. [3] compared DNNs against classical ML algorithms and showed that deep models can outperform classical classifiers given balanced data and proper feature engineering. Alatram et al. [4] contributed DoS/DDoS-MQTT-IoT, a dataset focused on volumetric and protocol-level denial-of-service evaluations.

Ullah et al. [5] proposed TNN-IDS, a transformer-based IDS that captures long-range sequence dependencies and demonstrates robust detection of evolving multi-stage attacks. Vijayan and Sundar [6] developed a hybrid DL pipeline with an autoencoder for unsupervised representation learning and an LSTM-based Deep Belief Network for sequence anomaly detection. Husnain et al. [7] promoted protocol-aware preprocessing via an MQTT parsing engine to reduce malformed-packet-induced false positives. Li and Ahmed [8] introduced XMID-MQTT, an explainable ensemble IDS/IPS with interpretable outputs for operator validation.

On the prevention side, Husnain et al. [7] and Syed et al. [9] demonstrated integrating protocol-aware filtering and fog–cloud prevention architectures, respectively. Zhang et al. [10] and Alotaibi et al. [9] explored federated and distributed ML architectures for privacy-preserving, scalable detection and prevention. Lightweight cryptographic and authentication schemes combined with ML for authentication were proposed by Bangare et al. [11]. Adaptive rate control methods to catch low-rate or stealthy DoS attacks were examined by Al-Fayoumi et al. [12]. Reviews and surveys [13]–[19] summarize broader IoT IDS trends including edge-based detection, hybrid CNN–RNN models, and federated frameworks.

Despite these advances, many studies still rely on generalized datasets (BoT-IoT, CICIDS2017, UNSW-NB15) lacking MQTT-specific semantics; this motivates the design of MQTT-aware datasets and lightweight, protocol-aware detection/prevention systems. Table 1 summarizes representative IDS/IPS works and datasets in MQTT/IoT research, including methodological class, dataset, attacks covered, reported performance metrics, strengths, and limitations.

Table 1. Summary of IDS/IPS Techniques and Datasets Used in IoT and MQTT-Based Networks.

Author (Year) / Ref	Main ML / DL Technique Used	Dataset Used	Attacks Addressed	Reported Performance Metrics	Key Strengths	Key Limitations
Vaccari et al. (2020) [1]	Classical ML (SVM, RF, DT)	MQTTset (simulated MQTT)	DoS, spoofing, malformed packets	Accuracy ≈ 97%, Precision ≈ 95%	First MQTT-specific benchmark dataset	Limited scale and attack diversity
Hindy et al. (2020) [2]	Classical ML (flow vs packet)	MQTT-IoT-IDS2020 (simulated)	DoS, spoofing, recon	FPR < 2%, Detection rate ≈ 98%	Showed flow-based features	Synthetic dataset; limited realism

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Author (Year) / Ref	Main ML / DL Technique Used	Dataset Used	Attacks Addressed	Reported Performance Metrics	Key Strengths	Key Limitations
					improve accuracy	
Khan et al. (2022) [3]	Deep Neural Network (DNN)	MQTTset / custom MQTT traffic	Anomaly, DoS, spoofing	Accuracy 99%, F1 ≈ 98%	Deep model outperforms ML on balanced data	High compute requirement; sensitive to imbalance
Alatram et al. (2023) [4]	Ensemble ML (RF, GBM, SVM)	DoS/DDoS-MQTT-IoT (simulated)	DoS, DDoS	ROC/AUC > 0.97	Focused DoS dataset; robust evaluation	Limited attack types (only DoS/DDoS)
Ullah et al. (2023) [5]	Transformer Neural Network (TNN-IDS)	Simulated MQTT sequences	Multi-stage & evolving attacks	Precision 99%, Recall 98%	Captures long-range dependencies via attention	High complexity; not edge-friendly
Vijayan & Sundar (2023) [6]	Hybrid DL (Autoencoder + LSTM DBN)	Simulated MQTT datasets	Sequential and anomaly attacks	F1 ≈ 97%, Detection ≈ 98%	Combines unsupervised + temporal learning	Complex tuning; requires large data
Husnain et al. (2022) [7]	Protocol-aware ML + Parsing Engine	Real MQTT captures	Protocol violations, DoS	FP reduction ≈ 35%	Reduces false positives using validation	Adds processing overhead; not full IDS
Li & Ahmed (2024) [8]	Explainable Ensemble ML (XMID-MQTT)	Mixed MQTT + simulated traffic	DoS, spoofing, anomalies	Accuracy ≈ 99%, Explainability score high	Interpretability + high accuracy	Extra compute for explainability
Alotaibi et al. (2024) [9]	Distributed ML (Edge/Fog Framework)	Simulated MQTT networks	Multi-node coordinated attacks	Real-time detection > 97%	Scalable + low latency architecture	Complex synchronization across nodes
Syed et al. (2023) [10]	ML at Fog/Edge Layer	Fog-Cloud testbed	DDoS, Brute Force IoT	Accuracy ≈ 98%	Edge-localized decision making	Bandwidth overhead for aggregation
Bangare et al. (2024) [11]	Lightweight Crypto + ML	Simulated MQTT auth flows	MITM, replay attacks	Latency < 5 ms; Accuracy > 95%	Secure hash-chain auth with low latency	Limited dataset validation
Al-Fayoumi et al. (2023) [12]	Statistical ML + Adaptive Thresholds	Custom MQTT flows	Low-rate DoS / DDoS	Detection rate ≈ 97%	Temporal adaptive rate control effective	May misclassify bursty legitimate traffic
Raza et al. (2017) [13]	Conceptual taxonomy (Signature/Anomaly/Hybrid)	N/A (Survey)	IoT attack taxonomy	N/A	Foundational lightweight IDS guidelines	No empirical evaluation
Ferrag et al. (2020) [15]	DL Survey (Stacked AE, CNN, RNN)	BoT-IoT, CICIDS, UNSW	Botnet, DDoS, injection	Comparative accuracy studies	Comprehensive DL analysis for IoT	Generalized datasets; not MQTT-specific
Aminanto et al. (2019) [17]	One-Class Classification (OC-SVM)	Simulated edge data	Local anomalies, DoS	Accuracy ≈ 94%	Edge-lightweight; low latency	Higher false positives on new attacks
Nguyen et al. (2023) [19]	Federated Deep Learning (CNN + LSTM)	Distributed IoT datasets	DDoS, scanning, malware	Accuracy ≈ 98%	Privacy-preserving training without raw data	Requires high sync and comm overhead
Koroniotis et al. (2019) [20]	N/A (Dataset Paper)	BoT-IoT	Botnet, DDoS, backdoor	N/A	Large-scale IoT testbed dataset	Synthetic traffic; not MQTT
Sharafaldin et al. (2018) [21]	N/A (Dataset Paper)	CICIDS2017	DDoS, brute force, web attacks	N/A	High-quality labeled flows	Generic network focus; not IoT
Moustafa & Slay (2015) [22]	N/A (Dataset Paper)	UNSW-NB15	Exploits, DoS, reconnaissance	N/A	Modern attack diversity	Requires protocol-level feature adaptation

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Author (Year) / Ref	Main ML / DL Technique Used	Dataset Used	Attacks Addressed	Reported Performance Metrics	Key Strengths	Key Limitations
Moustafa et al. (2020) [23]	Statistical + Feature Fusion	TON_IoT	Insider, DDoS, malware (IIoT)	Multi-metric comparisons	Real IoT + IIoT sensors included	No MQTT topics
Al-Harbi et al. (2022) [24]	ML (Ensemble RF + DL)	Edge-IIoTset	DoS, injection, malware	Accuracy ≈ 99%	Modern Edge/IIoT coverage	Complex feature extraction
Ullah & Mahmoud (2020) [25]	ML (Random Forest, DT)	IoTID20	DDoS, MITM, brute force	Accuracy ≈ 96%	Lightweight features; IoT focus	Limited real traffic
Meidan et al. (2018) [26]	DL (Autoencoder AE)	N-BaIoT (real IoT devices)	Botnet (Mirai, Bashlite)	Detection > 99%	Real IoT captures; time-series data	Not MQTT; specific botnets only

Dataset choice strongly affects IDS/IPS evaluation. Table 1 have details of major datasets: MQTTset [1], MQTT-IoT-IDS2020 [2], DoS/DDoS-MQTT-IoT [4], BoT-IoT [20], CICIDS2017 [21], UNSW-NB15 [22], TON_IoT [23], Edge-IIoTset [24], IoTID20 [25], N-BaIoT [26] and more recent contributions [27]–[34]. Many early datasets are general-network or IoT-general; MQTT-specific datasets are recent and often simulated or limited to particular attack types, reinforcing the need for diverse, multi-scenario MQTT benchmarks.

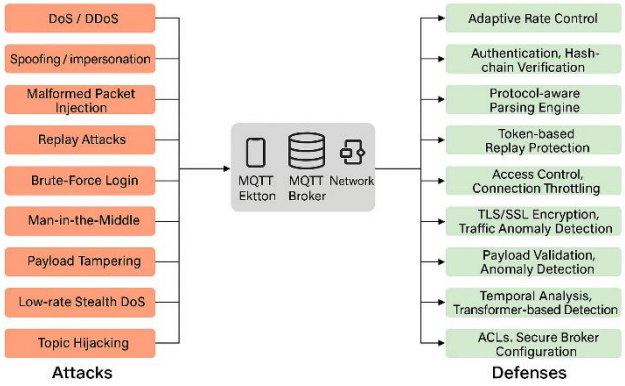


Fig. 3 Visualization of MQTT-Based Attacks and Corresponding Defense Mechanisms

Figure 3 illustrates the relationship between major MQTT-specific attack categories and their corresponding defense mechanisms as identified in recent studies. It highlights that most countermeasures focus on detection, while real-time prevention and adaptive resilience remain open research directions.

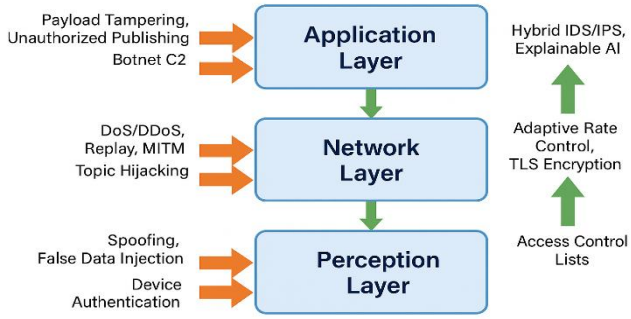


Fig. 4 IoT Layers Affected by MQTT-Based *Attacks*

The above figure illustrates the impact of MQTT-based attacks across different IoT layers. The diagram shows how threats propagate from the perception to the application layer, along with representative defenses used at each level



Fig 5. Dataset Comparison Overview: Evolution and Diversity in IoT and MQTT-Based Intrusion Detection Research.

III. Challenges, Open Issues and Research Gaps

Despite significant progress in MQTT-based intrusion detection and prevention research, several challenges remain unresolved, creating opportunities for further advancement. The following points summarize key limitations in existing literature and highlight the corresponding directions for improvement.

1. Limited and non-standardized MQTT datasets:

Existing MQTT datasets such as MQTTset, MQTT-IoT-IDS2020, and DoS/DDoS-MQTT-IoT remain small, scenario-specific, or simulation-based, reducing reproducibility and comparative evaluation across studies [1], [2], [4].

2. Resource constraints of IoT devices:

Deep learning–based IDS models provide strong accuracy but are often too computationally demanding for low-power IoT nodes, emphasizing the need for lightweight and hybrid architectures [3], [5], [17].

3. Real-time detection and latency limitations:

Many IDS models perform well under offline testing but fail to sustain low-latency, real-time detection in continuous MQTT traffic streams [5].

4. Scalability and evolving threat adaptation:

Centralized and static IDS/IPS designs scale poorly in large IoT deployments and are ineffective against emerging or zero-day attacks, highlighting the importance of federated and adaptive learning [9], [10], [19], [31].

5. Insufficient protocol-aware feature usage:

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Several studies treat MQTT traffic as generic network data, failing to incorporate MQTT-specific attributes such as QoS levels, topic structures, and session states, which can improve detection precision [7], [29].

6. Underdeveloped prevention-oriented mechanisms: Most research focuses on detection rather than automated mitigation, leaving real-time prevention insufficiently addressed [4], [9], [30].

7. Inconsistent evaluation and benchmarking: Current studies employ non-uniform evaluation metrics often focusing only on accuracy or F1-score while neglecting latency, scalability, and energy consumption, leading to inconsistent comparisons [1], [15], [27], [32].

IV. Conclusion

This review synthesizes recent advancements in intrusion detection and prevention for MQTT-based IoT networks. While numerous studies demonstrate the potential of ML, DL, hybrid, and federated approaches, practical deployment remains challenged by dataset limitations, resource constraints, real-time requirements, and the need for explainability. To bridge these gaps, future research should prioritize creating realistic MQTT datasets, designing lightweight and adaptive IDS/IPS models, integrating protocol-aware feature engineering, and ensuring model transparency and safety in automated prevention. The PhD work associated with this review aims to address these areas by proposing a hybrid, explainable IDS/IPS framework optimized for MQTT ecosystems.

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